

Lecture 2: Simple linear regression

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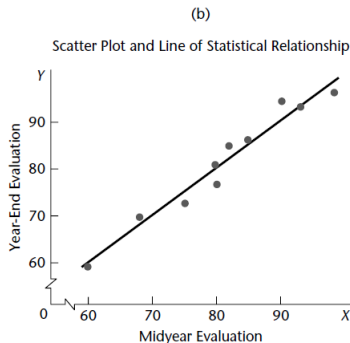
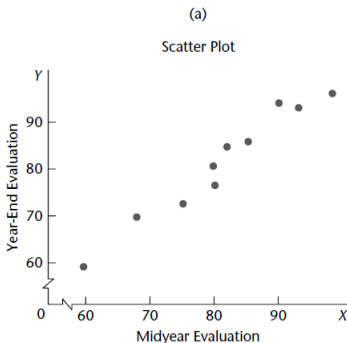
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 - Regression and model building
 - Data collection
- 2 Simple linear regression model
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- 7 Overview of steps in regression analysis

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Statistical relation between two variables

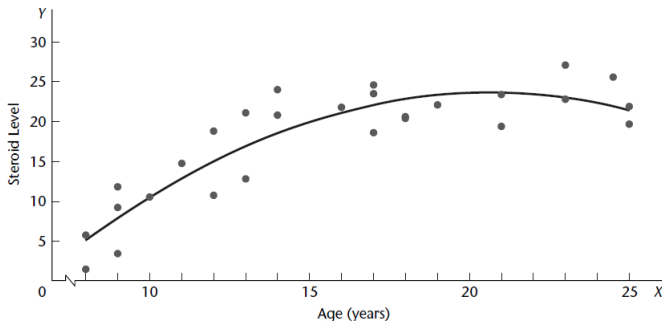
Example: Performance evaluations for 10 employees were obtained at midyear and at year-end.

- Year-end evaluations (Y) are the **dependent** or **response variable**
- Midyear evaluations (X) are the **independent/ explanatory/ predictor variable**.



Statistical relation between two variables (contd.)

Curvilinear statistical relation between age and steroid level in healthy females aged 8 to 25.



The term **regression** describe statistical relations between variables.

Regression model and their uses

Regression analysis is a **statistical technique** for investigating and modeling the relationship between variables.

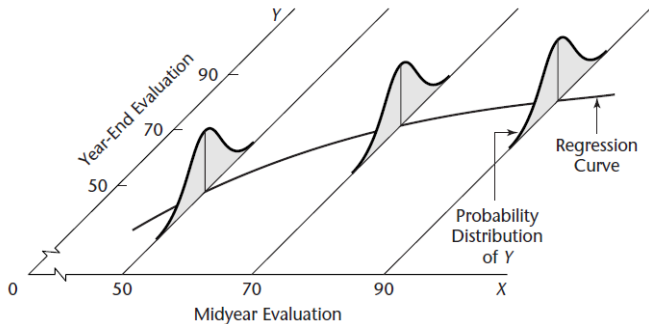
Regression models are used for several purposes

- Data description
- Parameter estimation
- Prediction and estimation
- Control

Regression models

Two essential assumptions in a regression model are:

- There is a probability distribution of Y for each level of X .
- The mean of this probability distributions is a function of X .



Regression and Causality

- The existence of a statistical relation between the response variable Y and the explanatory or predictor variable X **does not imply in anyway** that Y depends causally on X .
- Check out:
<http://www.tylervigen.com/spurious-correlations>.

Data collection

Three basic methods for collecting data:

- ① **A retrospective study based on historical data:** often offer limited amounts of useful information
 - Some of the relevant data often are missing.
 - The reliability and quality of the data are often highly questionable.
 - The nature of the data often may not allow us to address the problem at hand.
 - The analyst often tries to use the data in ways they were never intended to be used.
 - Logs, notebooks, and memories may not explain interesting phenomena identified by the data analysis.
- ② **An observational study**
 - Observe the process, only interact or disturb the process as much as is required to obtain relevant data.
 - With proper planning, these studies can ensure accurate, complete, and reliable data.
 - However, these studies often provide very limited information about specific relationships among the data.
- ③ **A designed experiment:** The best data collection strategy

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Formal statement of model

The **simple linear regression** model is stated as follows:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, i = 1, \dots, n. \quad (1)$$

- Y_i : the value of the response variable in the i th trial
- β_0 and β_1 are parameters
- X_i : the value of the predictor variable in the i th trial
- ϵ_i : a random error term with **zero mean** ($E\{\epsilon_i\} = 0$); **constant variance** ($\text{var}\{\epsilon_i\} = \sigma^2$); and ϵ_i, ϵ_j are **uncorrelated** ($\text{cov}\{\epsilon_i, \epsilon_j\} = 0$)¹.

¹**Covariance** represents the correlation between two random variables:

$$\text{cov}\{X, Y\} = E\{(Y - E\{Y\})(Y - E\{Y\})\}.$$

Important features of model

- 1 Y_i is a random variable with mean:

$$E\{Y_i\} = E\{\beta_0 + \beta_1 X_i + \epsilon_i\} = \beta_0 + \beta_1 X_i + E\{\epsilon_i\} = \beta_0 + \beta_1 X_i \quad (2)$$

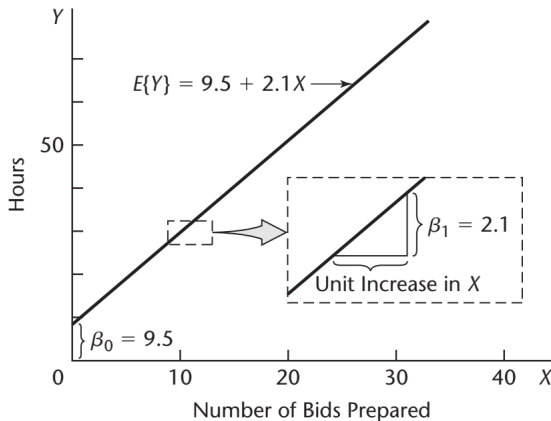
- 2 Y_i has constant variance:

$$\text{var}\{Y_i\} = \sigma^2,$$

where σ^2 is the variance of the error term.

Meaning of parameters of simple linear regression model

Example: A consultant for an electrical distributor is studying the relationship between the number of bids requested by construction contractors for basic lighting equipment during a week and the time required to prepare the bids.



Exercise: 1.7, page 33

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Least square method

- Given n observations $(X_1, Y_1), \dots, (X_n, Y_n)$.
- The deviation of Y_i from its expected value:

$$Y_i - (\beta_0 + \beta_1 X_i)$$

- The least square method minimizes the sum of n squares deviations (sum-square error):

$$SSE = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2. \quad (3)$$

Least square method

- Let b_0, b_1 are the optimal value of the least square problem minimizing SSE (a.k.a. (point) estimators of β_0, β_1).
- Using *first order condition* for problem (3) (partial derivatives of SSE with respect to β_0, β_1 equal to zero), we obtain

$$b_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$
$$b_0 = \bar{Y} - b_1 \bar{X}.$$

b_1 is a linear combination of the Y_i .

Properties of least squares estimators

Gauss-Markov theorem

Under the conditions of simple regression model, the least squares estimators b_0 and b_1 are unbiased and have minimum variance among all *unbiased linear estimators*^a.

^aUnbias means $E\{b_0\} = \beta_0$, $E\{b_1\} = \beta_1$.

Example: Toluca company

- page 19, [MPV]
- calculate the estimators
- double check with *o/s* function
- plot the results

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Run	Lot	Work					
i	Size	Hours	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$	$(Y_i - \bar{Y})^2$
1	80	399	10	86.72	867.2	100	7,520.4
2	30	121	-40	-191.28	7,651.2	1,600	36,588.0
3	50	221	-20	-91.28	1,825.6	400	8,332.0
...	...	...	...	...	...	...	...
23	40	244	-30	-68.28	2,048.4	900	4,662.2
24	80	342	10	29.72	297.2	100	883.3
25	70	323	0	10.72	0.0	0	114.9
Total	1,750	7,807	0	0	70,690	19,800	307,203
Mean	70.0	312.28					

Mean responses and residuals

- Mean response:

$$E\{Y\} = \hat{Y} = b_0 + b_1X \quad (4)$$

- Residual:

$$e_i = Y_i - \hat{Y}_i = Y_i - (b_0 + b_1X_i). \quad (5)$$

We have $\sum_{i=1}^n e_i = 0$.

	(1)	(2)	(3)	(4)	(5)
Run	Lot	Work	Estimated		
i	Size	Hours	Mean	Residual	Squared
	X_i	Y_i	Response	$Y_i - \hat{Y}_i = e_i$	Residual
			$\hat{Y}_i$		$(Y_i - \hat{Y}_i)^2 = e_i^2$
1	80	399	347.98	51.02	2,603.0
2	30	121	169.47	-48.47	2,349.3
3	50	221	240.88	-19.88	395.2
...	...	...	...	...	...
23	40	244	205.17	38.83	1,507.8
24	80	342	347.98	-5.98	35.8
25	70	323	312.28	10.72	114.9
Total	1,750	7,807	7,807	0	54,825

Properties of linear regression model

- 1 The sum of the residuals is zero: $\sum_{i=1}^n e_i = 0$.
- 2 The sum of the squared residuals, $\sum e_i^2$, is a minimum (least square).
- 3 The sum of the observed values Y_i equals the sum of the fitted values $\hat{Y}_i$: $\sum_{i=1}^n Y_i = \sum_{i=1}^n \hat{Y}_i$.
- 4 The sum of the weighted residuals is zero when the residual in the i th trial is weighted by the level of the predictor variable in the i th trial: $\sum_{i=1}^n X_i e_i = 0$.
- 5 The sum of the weighted residuals is zero when the residual in the i th trial is weighted by the fitted value of the response variable for the i th trial: $\sum_{i=1}^n \hat{Y}_i e_i = 0$.
- 6 The regression line always goes through the point $(\bar{X}, \bar{Y})$.

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Estimation of error terms variance

The variance σ^2 of the error terms ϵ_i in regression model needs to be estimated to obtain an indication of the variability of the probability distributions of Y .

- Error sum of squares or residual sum of squares (a.k.a. SS_{Res}):

$$SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n e_i^2.$$

- Error mean square or residual mean square:

$$s^2 = MSE = \frac{SSE}{n-2} = \frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{n-2} = \frac{\sum_{i=1}^n e_i^2}{n-2}.$$

SSE has $n - 2$ degrees of freedom associated with it. (Two degrees of freedom are lost because both β_0 and β_1 had to be estimated in obtaining the estimated means $\hat{Y}_i$.)

- Estimation of residual standard deviation $s = \sqrt{MSE}$.

Toluca example

Calculate SSE, MSE, the estimated standard deviation of Y in above Toluca example.

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Normal error regression model

- The least squares method provides unbiased point estimators of β_0 and β_1 that have minimum variance among all unbiased linear estimators.
- However, to set up interval estimates and make tests, we need to make an assumption about the form of the distribution of ϵ_i .
- The standard assumption is that the error terms ϵ_i **are normally distributed** which greatly simplifies the theory of regression analysis.

Normal error regression model

The **normal error regression model** is as follows:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i,$$

where

- Y_i is the observed response in the i th trial
- X_i is a known constant, the level of the predictor variable in the i th trial
- β_0 and β_1 are parameters
- ϵ_i are independent $N(0, \sigma^2)$
- $i = 1, \dots, n$.

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Single population

- Consider a normal population whose standard deviation is known to be $\sigma = 10$ and whose **mean is unknown**.
- A random sample of $n = 3$ observations is selected from the population and yields the results $Y_1 = 250$, $Y_2 = 265$, $Y_3 = 259$.
- We now wish to ascertain which value of μ is most consistent with the sample data.

Maximum likelihood

- The likelihood function of above sample with $\sigma = 10$ is

$$\begin{aligned} L(\mu) = & \left(\frac{1}{\sqrt{2\pi}10} \right) \exp \left[-\frac{1}{2} \left(\frac{250 - \mu}{10} \right)^2 \right] \\ & * \left(\frac{1}{\sqrt{2\pi}10} \right) \exp \left[-\frac{1}{2} \left(\frac{265 - \mu}{10} \right)^2 \right] \\ & * \left(\frac{1}{\sqrt{2\pi}10} \right) \exp \left[-\frac{1}{2} \left(\frac{259 - \mu}{10} \right)^2 \right] \end{aligned}$$

- Maximum likelihood estimation method is to find $\hat{\mu}$ to maximize $L(\mu)$.
- Estimator: $\hat{\mu} = \bar{Y} = 258$.

Regression model

- Each Y_i observation is normally distributed with mean $\beta_0 + \beta_1 X_i$ and standard deviation σ .
- Suppose we already know that $\sigma = 2.5$, the likelihood value for the parameter values $\beta_0 = 0$ and $\beta_1 = .5$ of the sample is

$$\begin{aligned} L(\beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^n \frac{1}{(2\pi\sigma^2)^{1/2}} \exp \left[-\frac{1}{2\sigma^2} (Y_i - \beta_0 - \beta_1 X_i)^2 \right] \\ &= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2 \right] \end{aligned}$$

Maximum likelihood estimators vs. least-square estimators

- $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\sigma}^2$ are the maximum likelihood estimator of β_0 , β_1 , and σ^2 .
- It is proved that maximum likelihood estimators $\hat{\beta}_0$, $\hat{\beta}_1$ are also the least-square estimator b_0 , b_1 founded previously

$$b_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$b_0 = \frac{1}{n} \left(\sum_{i=1}^n Y_i - b_1 \sum_{i=1}^n X_i \right) = \bar{Y} - b_1 \bar{X}.$$

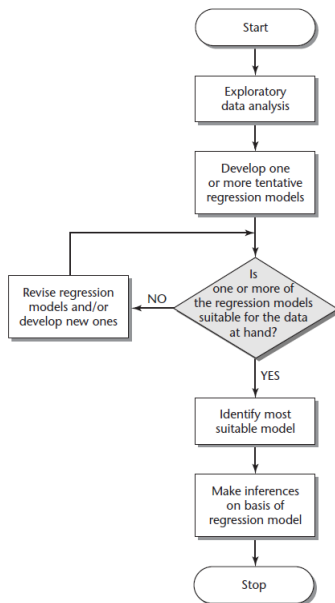
- Maximum likelihood estimator of the variance is

$$\hat{\sigma}^2 = \frac{\sum (Y_i - \hat{Y}_i)^2}{n},$$

which is an biased estimator of variance. (Note that the unbiased estimator $MSE = s^2 = \frac{\sum (Y_i - \hat{Y}_i)^2}{n-2}$.)

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Overview of steps in regression analysis



Summary

- X is predictor, Y is response
- Simple linear regression model: $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, i = 1, \dots, n.$
- Steps in regression analysis
- Estimation of regression function
- Estimation of error terms variance
- Maximum likelihood method vs. least square method

- M. Kutner, C. Nachtsheim, and J. Neter, Applied Linear Regression Models, chapter 1.
- MPV, Introduction to linear regression analysis, Chapters 1, 2.